



Health and Socioeconomic Status: The Importance of Causal Pathways

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The poor tend to be in poor health. A positive association between socioeconomic status (SES) and health has been established worldwide, within specific societies, over the life course, and for a broad array of indicators of both health and SES. Experimental and nonexperimental evidence on the causal pathways that underlie this association suggests that fetal health and health in early life likely affect health and economic prosperity in later life. Some evidence suggests that changes in health during adulthood also affect work outcomes. Overall, the strongest evidence for a causal impact of health on economic success is reflected in the effect of improved nutrition on economic productivity. The evidence on other dimensions of health is more limited. Very little attention has been paid to the dynamic interplay between health and behavioral choices that shapes economic outcomes over the longer term. Recent innovations provide unparalleled opportunities for making major contributions to knowledge that could have a lasting impact on the well-being of populations around the world.

The poor tend to be in poor health. A positive association between socioeconomic status (SES) and health has been established across the globe, within specific societies, over the life course, and for a broad array of indicators of both health and SES. Interpretation of the association is at the center of a large and contentious debate.

A key issue in this debate has been isolation of the causal pathways that link health and economic prosperity. Pinning down causal pathways at the individual level is critically important from both a scientific and a policy standpoint. For example, if income growth leads to improvements in health, policy makers should take health benefits into account when allocating resources to programs that promote economic growth.

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If, conversely, health contributes to economic prosperity, these benefits ought to be included in evaluations of the impact of resources allocated to the health sector—including, for example, provision of public health services, immunization, maternal and child health programs, subsidized medical care, and health insurance programs, as well as public health information campaigns and training of health personnel.

Causality likely runs in both directions. On the one hand, a family with more income is likely to invest more in its members' human capital, including health. In low-resource populations, as income increases, individuals typically invest in better diets and improved sanitation and devote more resources to health care. On the other hand, if a healthier worker is less susceptible to disease, more alert, and more energetic, he or she will probably be more productive and may work longer hours. As a result, the worker will command higher earnings. In contrast, individuals in poorer health may reduce their hours or be unable to work at all, and they may have to dip into their savings to pay for costly treatments or care.

Other, unobserved factors probably drive both health and economic prosperity. For example, part of health is inherited, as are other attributes related to economic prosperity. The role of genes, and the role the environment plays in their expression, are not fully understood. Moreover, families that choose to invest in the health of their children may also value other investments that would improve their own and their children's lives. They may make investments in other dimensions of their children's human capital, working more effectively, or saving more.

It is important to take into account factors that are not readily observed (or, more precisely, not treated as observed, in most models), for at least two reasons. First, a demonstration that health does not have a causal impact on income cannot be interpreted as indicating that the correlation between income and health should be attributed to a causal effect of income on health. The association may be attributed to these other, unobserved factors.

Second, the literature has typically referred to unobserved factors, including, for example, tastes, information, and the like. Some of these, such as information about health practices or propensities to save for the future, are amenable to measurement. Accumulation of scientific evidence on their role, if any, in behavioral choices that explain the association between health and economic prosperity is a potentially important new line of research. Moreover, some of these factors can be influenced by policy. For example, campaigns that provide information about health and the health consequences of behaviors have been successful in effecting lifestyle changes that promote better health, such as policies that discourage tobacco use.

This paper focuses on key issues involved in isolating causal pathways that link health and SES at the micro level. Rather than attempt exhaustively to describe the methods and evidence, we focus on a small number of studies to illustrate the methods that have been used in the literature and assess their contributions to knowledge. The conclusion highlights some of the lines of future inquiry in this literature that are likely to be especially profitable and describes some methodological advances that have the potential to yield new evidence that will substantially influence both science and policy design. (For more comprehensive reviews of the literature on health and development, see Strauss and Thomas 1995, 2008; Jack and Lewis 2008.)

Measurement of Health and Socioeconomic Status

The relationship between health and SES is not unidimensional. The magnitude and direction of the correlation vary with the measures of health and of SES. For example, short-term episodes of illness that have no longer-term ramifications for health or functioning may result in changes in how time is allocated (to rest and recuperation, say) and thus perhaps to earnings during the period of illness, although even this will depend on the nature of the work and the payment mechanisms. Workers paid by the piece or on an hourly or daily basis are likely to have lower earnings when sick, whereas workers with paid sick leave may not. Reallocations of time in response to episodes of illness may be temporary, and the individual may make up for lost earnings by spending more time on activities such as work when his or her health improves.

More persistent or chronic health conditions may have broader effects on SES. These conditions may result in permanent changes in earnings because of changes in time allocation or productivity, or both. The health conditions may also result in changes in consumption, savings, and wealth as individuals draw on resources for treatment or in response to reduced earnings. Conditions that affect life expectancy likely affect intertemporal preferences and therefore decisions about savings and wealth accumulation.

In practice, in many instances, whether a specific health event will turn into a chronic condition may not be immediately obvious, and the path from short-term to longer-term health difficulties may be complex. Moreover, the cumulative effect of multiple health insults may turn an apparently short-term health problem into a chronic condition. For example, the literature suggests that the deleterious consequences for day-to-day functioning of a traumatic event such as an earthquake or exposure to a violent attack are dramatically magnified when an individual has had prior exposure to traumatic events.

The impact of a health condition may also have long-reaching effects on economic and social success if the condition results in a shift in the individual's lifetime trajectory. For example, a health condition during early childhood may affect school performance, which, in turn, has lasting effects on academic achievement and economic success in later life. In other instances, behavioral responses to chronic conditions, such as change of occupation or life style, may mitigate the negative consequences for economic prosperity. Health innovations in later life can have yet other impacts on the economic security of the individual.

It is important to keep in mind that there is likely to be considerable heterogeneity not only in the magnitude and direction of the association between health and SES but also in its interpretation. These issues have received inadequate attention in the literature, partly because relatively few health measures have been used in studies relating health to SES, a shortcoming that reflects data constraints.

In recent years, the quality and quantity of data for exploring these questions have dramatically improved. There have been, and continue to be, exciting innovations in the technology for measuring a broad array of health markers, using noninvasive and minimally invasive methods that can be implemented in the field at low cost. These

technologies include, for example, in-home monitors and assays of blood drops (dried on filter paper), cheek swabs, saliva, and hair, for measuring short-term and longer-term health conditions as well as DNA. These health measures have been integrated with rich administrative data, as well as with population-based socioeconomic data, many of which involve long-term longitudinal surveys that follow individuals over the life course.

In combination, these integrated data sources provide opportunities to make major advances in understanding the complex interplay between different dimensions of health and SES over the life course. In the next sections, variation over the life course will be used as an organizing principle in an assessment of the state of the field regarding the causal impact of health on socioeconomic prosperity.

Health in Early Life and Later Outcomes

Investigation of the links between health in early life, including fetal health, and later life outcomes is an active area of scientific inquiry. Research by Barker and his colleagues has highlighted the key role of maternal nutrition during pregnancy. It is thought that inadequate nutrients and oxygen during fetal development may result in adaptations in organ development to ensure survival while a fetus and possibly immediately after birth. This fetal “programming” appears to result in elevated risk of health problems throughout the entire life course (Barker 1994; Victora et al. 2008).

For example, low birthweight may result in reduced growth during infancy because the body is programmed as a fetus to survive on low levels of nutrients. This “thrifty phenotype,” which develops as an adaptation to adverse nutrition in utero, results in permanent metabolic and endocrine changes. These changes would be beneficial if food continued to be scarce after birth, but if food is plentiful, the same metabolic changes may be disadvantageous; they have been associated with elevated risks of obesity, dyslipidemia, and glucose intolerance in later life (Hales and Barker 1992). Thus, it is thought that being born small and thin but becoming overweight during adolescence or adulthood put an individual at greatest risk of heart disease and type 2 diabetes in later life. Furthermore, the biological literature suggests that not only does the overall nutritional status of the mother affect fetal health; the timing of nutrition and health insults also affect the growth and development of the child. For example, inadequate nutrients during the period when arteries are formed are associated with elevated risk of heart disease because of hardened arteries in midlife and later (Godfrey and Barker 2000).

Much of the experimental evidence on the longer-term effects of maternal nutrition deprivation relies on animal models. There are, however, several innovative studies that exploit quasi-experimental designs to measure the effects on humans. For example, Roseboom et al. (2001) summarize a rich body of work that examines the impact of the Dutch famine in the Netherlands on the health and mortality of prime-age adults. Toward the end of World War II, several Dutch cities that were thought to be supporting the Resistance were subjected to severe food rationing for about six months. People who were in utero during that period constitute a “treatment group”

whose health in later life can be compared with that of a “control group” of people who were born at the same time in other Dutch cities and were not subjected to the same food rationing. The authors find that nutritional deprivation early in the fetal period results in an elevated risk of glucose intolerance and heart disease among adult males and females.

It is likely that maternal nutritional status is not the only factor that affects fetal outcomes. Finch and Crimmins (2004) highlight the importance of infection during the fetal period. They argue that maternal infection and stress result in inflammation that has detrimental consequences for the development of the fetus, which affects the health and well-being of the offspring through the entire life course. For example, it is thought that inflammation in utero is implicated in the development of plaque in the arteries and therefore in heart disease and that these effects are independent of diet and lifestyle choices.

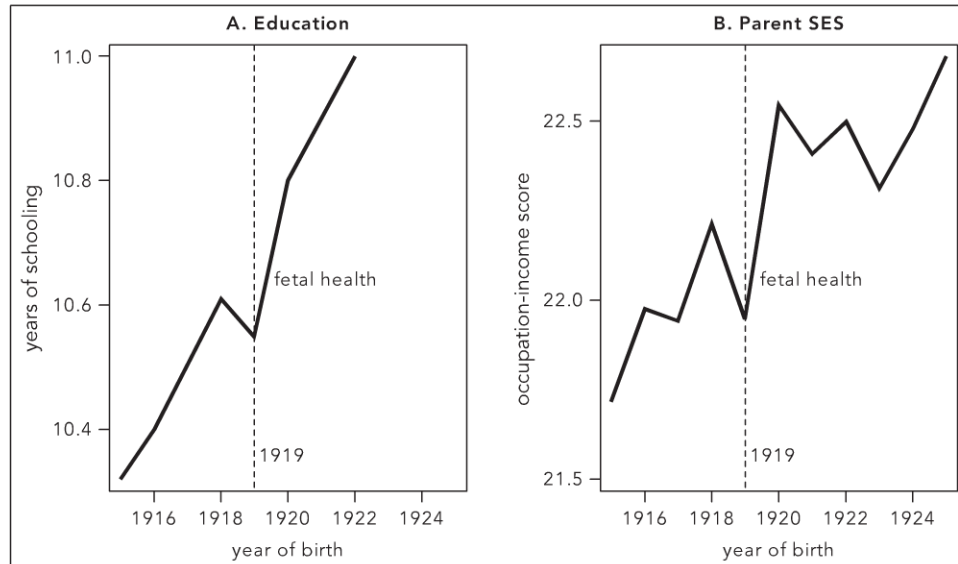
Quantification of the magnitude of these effects is particularly salient for developing countries because of the coexistence of infectious diseases with the onset of the nutrition transition. On the one hand, the obesity epidemic is reaching beyond the developed countries and is spreading throughout the globe (Popkin 1993). On the other hand, levels of infection and, thus, of inflammation are much higher in poorer societies, suggesting that these populations may be carrying a particularly heavy burden of disease in the future.

A key implication of this research is that the long arm of fetal and early childhood health may extend not only to health in later life but also to economic and social prosperity through the life course. Studies have documented that low birthweight is predictive of lower school attainment and less success in the labor market. Although low birthweight likely reflects poor intrauterine growth, it is not clear that these associations can be given a causal interpretation. Low birthweight is the outcome of many factors, including low resource levels, inadequate access to health services, and maternal behaviors, all of which may be related to other investments in infants and children. To establish that variation in fetal health causes variation in economic success, it is necessary to identify changes in fetal health that are arguably exogenous and not under the direct control of parents.

This concern is addressed in some recent studies. Almond (2006) examines the longer-term effects of being conceived in late 1918, during the Spanish flu pandemic. The Spanish flu, caused by an unusually severe strain of the H1N1 virus, resulted in very high levels of morbidity and mortality. It is estimated that more than half a million people died in the United States and that about one in four people suffered from the virus (Crosby 1989). The virus was first detected in March 1918, and although the pandemic persisted to the middle of 1920, its impact on the health of the U.S. population was concentrated in the last half of 1918. Whereas influenza usually kills the youngest and oldest, the Spanish flu killed prime-age adults and fell particularly heavily on males. It has been conjectured that undetected tuberculosis among these men rendered them particularly vulnerable (Noymer and Garenne 2000).

Using data from the 1960 United States census, Almond compares the health, schooling, and labor market outcomes of people who were born in 1919 with those born before the onset of the pandemic and those born at least nine months after the

FIGURE 1.
Fetal Health, Own Education, and Socioeconomic Status (SES) of Parents, by Year of Birth, United States



Source: Panel A, 1960 U.S. Census; panel B, 1930 U.S. Census.

Note: The 1919 cohort was age 41 at the time of the 1960 census (panel A) and age 11 at the time of the 1930 census (panel B).

pandemic ended. As shown in panel A of figure 1, he finds that, relative to other cohorts, males who were born in 1919 (and were thus in utero during the most virulent phase of the pandemic) completed significantly fewer years of schooling. They are also less likely to have graduated from high school, and at age 40 they earned less than control cohorts.

Almond attributes these effects to the impact of maternal infection on the fetus. It is possible that they reflect other influences such as higher prices for food and other goods at the time of the pandemic, lower incomes because of reduced labor supply at the time of the pandemic, or stress associated with the pandemic. It is also possible that the results reflect the impact of the pandemic on childbearing decisions of parents. The influenza virus emerged in early 1918, when soldiers were returning from World War I. Parents who chose to have children at that time may have been different from those who had recently given birth and those who delayed childbearing.

To investigate this issue, panel B of figure 1 presents an indicator of SES of parents for each cohort of births in the 1930 census. The SES index is based on an occupation-income score. It is significantly lower for children born in 1919 than for those born in 1920, suggesting either that lower SES parents were more likely to have children in 1919 than higher SES parents or that children of higher SES parents in the 1919 birth cohort were more likely to have died by age 11. The latter seems unlikely. There is a strong intergenerational correlation between education and SES, and so the fact that the 1919 birth cohort has less schooling in adulthood reflects, at least in part,

differences in family background. It would seem premature to attribute the entire gap between the 1919 and other birth cohorts in economic prosperity as an adult to the impact of health insults during the fetal period resulting from the influenza pandemic. Almond also exploits variation across states in mortality attributed to the pandemic, but those estimates are potentially contaminated by behavioral responses to the pandemic. (See Thomas 2008 for a discussion.)

Related work examines the impact on human capital and economic prosperity in midlife of being born during the 1959–61 Chinese famine. Using the 1991 wave of the China Health and Nutrition Survey (CHNS), Chen and Zhou (2007) compare outcomes for cohorts born in the 1950s and 1960s. The impact of the famine is identified by using regional variation in the 1960 death rate that is attributed to the famine. The authors find that adults who were born in 1959 or 1960 and survived to 1991 would have been 3 centimeters taller if they had not been exposed to the famine. They also report reduced hours of work by individuals in this cohort. Family income per capita is lower, but the magnitude and significance of these effects differ across the cohort, depending on the specific measure of income. It is unclear how to interpret evidence on family earnings, which reflect the contributions of family members from other birth cohorts.

As with the influenza pandemic, it is possible that the estimated effects of the Chinese famine reflect the impact of income and price variation resulting from the famine. As the authors note, interpretation of the estimates is complicated by at least two additional concerns. First, location of birth is not known in the CHNS, and so the authors assume that respondents were born in the same location where they were interviewed 40 years later. This, they conjecture, is a reasonable assumption in China during the period. This concern highlights an important advantage of Almond's work, which uses census data that include information on location of birth.

Second, survivors of the famine may be selected on health and human capital. On the one hand, it is likely that the frailest individuals would have died in the harsh environment at the time of the famine—in which case the estimated effects are probably downward biased. On the other hand, there is evidence that fertility responded to the famine and that birth rates declined substantially soon after the onset of the famine. The direction of bias in this case is unclear.

One assessment of the empirical magnitude of these influences comes from a comparison of birth cohorts born around the time of the 1974 floods in Bangladesh with cohorts born before and after the floods. Flooding is commonplace in the Ganges Delta, but there had been no parallel to the severity of the 1974 floods for almost 20 years, and it is estimated that more than 20,000 people were killed by them. Table 1 presents data from the 1996 Matlab Health and Socioeconomic Survey (MHSS), which collected population-representative information on individuals living in Bangladesh's Matlab District. The fraction of the population born between 1970 and 1978 is reported in the first column for each year of birth. The substantially reduced size of the 1975 birth cohort is consistent with the idea that fetal health was affected by the floods. The second column reports differences in the fractions of males in each birth cohort who are unable to read, relative to the 1970 birth cohort. Cohorts born in 1975 are nearly 10 percentage points less likely to be illiterate than those born in

the early 1970s or in the late 1970s. These are very large differences, given that about 16 percent of males born in the 1970s are unable to read. The results are very similar when we compare males who are both illiterate and innumerate, in the third column. (There is no evidence of a secular trend in literacy rates during the 1970s. By contrast, comparisons of school attainment and height are complicated by secular trends.)

The fact that the cohort of males exposed to the floods has higher levels of human capital suggests that these estimates are dominated by selection rather than by the deleterious impact of the floods on fetal health. Two additional pieces of evidence suggest that this is a legitimate interpretation of the evidence. First, the 1974 and 1976 birth cohorts are also more likely to be literate and numerate than cohorts born before or after them. The 1976 cohort was not directly exposed to the flood.

Second, by reestimating the models with a household fixed effect, it is possible to assess the extent to which families with children who were born between 1974 and 1976 and who survived to 1996 are selective on characteristics that are associated with investments in human capital. The results are shown in the last two columns of table 1. The

TABLE 1. Human Capital and Exposure to the 1974 Floods in Bangladesh: Males, by Birth Year

Year of birth	Percent of population (1)	OLS		Household fixed effects	
		Percent illiterate (2)	Percent illiterate and innumerate (3)	Percent illiterate (4)	Percent illiterate and innumerate (5)
1971	9.2 (0.5)	0.9 (0.2)	1.1 (0.3)	-3.2 (0.6)	-2.8 (0.5)
1972	9.4 (0.5)	-1.7 (0.5)	-2.5 (0.7)	-0.4 (0.1)	-0.5 (0.1)
1973	11.7 (0.5)	-1.9 (0.6)	-2.6 (0.8)	-6.1 (1.2)	-5.7 (1.2)
1974	11.4 (0.5)	-6.3 (1.8)	-7.2 (2.2)	-3.8 (0.7)	-3.3 (0.7)
1975	9.0 (0.4)	-9.6 (2.6)	-9.6 (2.8)	-5.2 (0.9)	-4.8 (0.9)
1976	13.0 (0.5)	-8.4 (2.5)	-9.0 (2.9)	-3.0 (0.6)	-3.7 (0.8)
1977	14.4 (0.5)	-0.2 (0.0)	-0.2 (0.1)	-0.2 (0.0)	1.7 (0.4)
1978	12.6 (0.5)	-1.1 (0.3)	-0.6 (0.2)	0.5 (0.1)	0.6 (0.1)
Constant		16.0 (6.1)	14.8 (6.0)	15.3 (3.9)	13.4 (3.7)

Source: Author's calculations using data from the Matlab Health and Socioeconomic Survey on respondents born between 1970 and 1978.

Note: OLS, ordinary least squares. Number of males is 2,064. Numbers in parentheses are t-statistics. Birth year 1970 is the excluded category in regressions reported in columns (2) through (5).

differences in human capital across birth-year cohorts are substantially reduced, and there is no evidence that the 1974–76 cohorts are different from the others. We conclude that at least in the case of the Bangladesh floods, inferences about the causal effect of health insults during the fetal period need to take into account behavioral responses by parents to the floods and their aftermath.

Selective fertility, and fetal or infant mortality that is selective on characteristics related to investments in health and human capital, are potential concerns for the interpretation of results based on comparisons across cohorts. In the Chinese case, there is evidence that the number of live births declined dramatically soon after the onset of the famine. Evidence based on children born in the United States in 1919 is also potentially contaminated by selective fertility. The general point is that individuals, couples, and families are likely to take into account changes in the environment, including changes in the disease environment, when making decisions about the future.

The emerging body of literature on the health and economic impact of insults in utero is extremely important. A related line of inquiry assesses the extent to which investments in the health and well-being of reproductive-age women, especially pregnant women, contribute to the health and economic well-being of the next generation. There are other potential benefits from investments in the reproductive health of women. Greater control over fertility typically leads to fewer children ever born and, possibly, to greater investments in each child (Schultz 2008). Moreover, there may be benefits in the form of better health of reproductive-age women. Greater control over timing of births and improved health potentially improve labor market opportunities for women and the resources available to them. To the extent that these factors also elevate women's control over resources within a family, the literature suggests that further investments in the health and well-being of children, as well as in the accumulation of wealth, may result (Thomas 1990; Rubalcava, Teruel, and Thomas 2009).

These issues are explored in a series of studies that use observational data and exploit variation in the availability of reproductive health services across Indonesia. The *bidan desa*, or village midwife program, inaugurated in the mid-1990s, seeks to place a trained nurse in every village in Indonesia in an effort to reduce maternal mortality and improve birth outcomes. The cost of the midwife is absorbed by the government for the first three years, as long as the midwife remains working in the village to which she is assigned. During this time, she is encouraged to develop a private practice.

Evaluation of the program is not straightforward. Comparisons of the health of women before and after a midwife joins a community would reflect both the impact of the midwife and any other changes over time. From the outset, midwives were allocated to the poorest communities that did not already have a midwife. Comparisons of the health of females in a community that had a midwife with the health of females in a community without a midwife would therefore reflect the combined influence of the presence of the midwife and differences between the communities.

To identify the effects of the midwives, Frankenberg and Thomas (2001) suggest that if midwives focused on the health of reproductive-age females and had no direct

influence on the health of males, differences between communities would be reflected in the health of males. Drawing on two waves of the Indonesia Family Life Survey (IFLS), they estimate the change in health of females in communities that did not have a midwife by the second wave and compare this change with the change among males in the same communities. This method provides an estimate of health changes in those “control communities.” The difference in changes in health of females and of males in communities in which a midwife joined the community between the survey waves provides an estimate of the differential health changes in “treatment communities.” Comparison of the differences in treatment communities with the differences in the control communities yields an estimate of the impact of the village midwife on the health of reproductive-age females. This estimation strategy takes into account the fact that midwives are allocated purposively to communities where they are likely to have their biggest impact. The strategy also takes into account secular changes in health within each community, as long as those changes are common for males and females. Frankenberg and Thomas (2001) focus on body mass index (BMI), an indicator of general health and nutritional status, and report a large and significant beneficial impact on the BMI of females from the introduction of a midwife into a community.

The strategy assumes that the midwife has no direct or indirect impact on the health of males. Indirect benefits may arise if women learn about better health care practices that affect other family members or if women seek advice from the midwife about the health of others in their families. If these benefits are positive, the estimated effects of the midwife will be biased downward. If, however, the presence of a midwife results in reduced resources being allocated to improving the health of males and children, the strategy would yield upward-biased estimates.

Frankenberg, Suriastini, and Thomas (2005) address this concern by examining the impact of a midwife on child health. In this case, the identification strategy combines the advantages of the longitudinal design of the IFLS with insights from the nutrition literature on the biology of linear growth. Specifically, health and nutrition interventions have their greatest impact on the length of children during the first few years of life. Eventual adult height is largely determined by around age 3 or 4. Thus, if a midwife joins a community when a child is age 5 or 6, the services provided by the midwife are unlikely to have much effect on the child’s growth trajectory. In contrast, if the midwife is present from the time the child is conceived, the child will be fully exposed to the potential benefits from the midwife.

Consider a treatment community in which a midwife joins the community immediately after the first wave of the survey. Comparison of the height of children age 8 in the first wave with children the same age in the second wave provides an estimate of the changes in height that can be attributed to secular improvements and are unlikely to be associated with the presence of the midwife. In contrast, the differences between the heights of children age 3 and younger in the second wave and of children of the same age in the first wave provide an estimate of the impact of being fully exposed to the midwife relative to having no exposure to her. This difference, however, is contaminated by secular changes that might have occurred during this period. Thus, comparison of the differences in heights of young children between the waves

with the differences in heights of older children between the waves (which capture secular changes) yields an estimate of the impact of the midwife on the longer-term nutritional status of young children.

It is possible that there are cohort-specific growth trajectories. To assess whether this is an important concern, an alternative strategy compares changes in heights of young children, relative to older children, in treatment communities with similar changes in heights of young and older children in control communities that never had a midwife. Furthermore, by comparing changes in the heights of older children in treatment and control communities, it is possible to evaluate the validity of the assumptions in the model. A child who is fully exposed to a midwife is almost half a standard deviation taller than a similar child who is not so exposed—a difference that is not only significant but is larger than the urban-rural gap in heights of Indonesian children. Related research shows that midwives are associated with improved birth outcomes, including increased birthweight and fewer complications at birth (Frankenberg et al. 2009). Taken as a whole, this research indicates that the introduction of improved reproductive health services in rural Indonesia has resulted in improved health of women and children.

These studies, based on observational data, are able to estimate effects that can be interpreted as causal because of the research designs, which develop credible control groups with which treatments are compared. Assessment of the impact of improved reproductive health services on economic outcomes is more tenuous with this strategy, although the evidence suggests that improved reproductive health services are associated with elevated rates of labor force participation in the market and with higher earnings among females in Indonesia.

Joshi and Schultz (2007) make further progress on this front by examining the impact of a family planning intervention that was assigned to some villages in Matlab District, Bangladesh, and not to others. Starting in 1977, family planning services, in conjunction with maternal and child health services, were provided on a door-to-door basis in treatment villages. Control villages continued to receive services at local clinics. Studies have documented large and significant declines in fertility (ICDDR,B 2004). Joshi and Schultz exploit this design to assess the impact on the health and economic prosperity of women exposed to the treatment and to measure the effects of exposure to the program on the women's children.

Drawing on survey data collected by the Matlab Health and Socioeconomic Survey (MHSS) 20 years after the initiation of the program, Joshi and Schultz report significant improvements in the health of exposed women, higher rates of employment out of the home, and greater economic productivity in those activities. The households in which these women reside are wealthier and are more likely to have access to water within the family compound. The children of exposed women are more likely to survive to age 5, and they are more likely to be immunized. There is some evidence of improved nutritional status among these children, although there are only modest gains in schooling relative to children of women in control areas.

Frankenberg and Thomas (2007) highlight the potential impact of exposure to the program on female savings and empowerment. Specifically, they exploit a feature of the MHSS, which, following the IFLS, collects information on assets at the individual

level within each household. Reduced fertility associated with exposure to the family planning outreach will reduce the number of children that a woman can call on for assistance at older ages, and so there is a greater incentive to accumulate wealth over the life course to draw on in later life. Moreover, if control over assets translates into control over decisions about resource allocations, as women gain control over fertility they may also be empowered to assert themselves in other domains of the family economy. Women who were exposed to the family planning outreach treatment have significantly higher asset levels than women not in the treatment areas. Treated women tend to accumulate gold and goats—assets that are traditionally controlled by women. This is not because couples in treatment areas accumulated more assets, since males in the treatment communities did not save more than males in control communities.

The combination of data from experiments and from observational sources paints a picture in which reproductive health programs yield benefits measured not only in increased control over fertility but also in improved health and economic outcomes for women and greater accumulation of resources for the future. Furthermore, there is some evidence that the programs yield benefits in terms of improved child health. We turn next to a discussion of the evidence on the influence of child health on economic prosperity.

Child Health and Economic Success

A substantial body of literature has documented that adult height is correlated not only with improved health but also with economic prosperity, as indicated by higher rates of participation in the labor force and, among those who work, higher hourly earnings. These relationships have been established in the historical literature following pioneering work by Fogel (2004) and his collaborators, and they have also been documented in the literature on developing countries (Strauss and Thomas 1998). Adult height is largely determined during early childhood, and clearly, economic prosperity during adulthood cannot have a causal impact on linear growth during childhood. It is therefore tempting to interpret these associations as indicating a causal effect of early childhood nutrition on economic prosperity during adulthood.

That would be premature. Factors that affect early childhood growth, such as genes, parental resources, and parents' tastes for investing in the human capital of their children, might also affect economic prosperity during adulthood. In that case, height and educational attainment, and possibly cognitive achievement, would be positively correlated (Case and Paxson 2008). Genetic influences, cognition, and tastes are seldom included in empirical models that relate adult height to economic prosperity, but without taking them into account, it is difficult to interpret the associations between attained adult height and economic prosperity.

Observational studies that relate height or weight of children to school performance suffer from similar problems, although attempts have been made to treat the joint determination of child health and child schooling through an instrumental variables approach or by including family fixed effects in the models. Prices of inputs into

the production of height are potential instruments; they may include (implicit or explicit) prices of local health services and prices of specific foods, relative to other prices. Estimation is complicated because child nutrition reflects both a stock and a flow component, and so the instruments necessarily capture variation in the environment to which the person was exposed during the fetal period and in the first few years of life. As noted above, the use of local variation in prices and services is complicated if allocation of programs and subsidies takes into account child health or if people move to areas because of improved services (Schultz 2002).

Alderman, Hoddinott, and Kinsey (2006) use variation in resources during the first years of life to measure the impact of early childhood nutrition on adult stature and schooling attainment. They exploit variation in resources resulting from droughts and civil war in Zimbabwe, treating this variation as a result of unanticipated shocks. It is not clear, however, that this variation is entirely unanticipated and that parents do not respond to it by, for example, moving.

The authors find that early childhood nutrition has a large and significant impact on years of completed schooling, indicating that public health interventions that ensure that infants and young children are well nourished are likely to have a direct impact on the human capital and economic prosperity of a population. (See Glewwe and Miguel 2008 for a review of this literature.)

Numerous animal experiments and some small-scale randomized interventions with children have implicated both macronutrients and several micronutrients such as iron and zinc in cognitive functioning. Some of these studies have linked supplementation to school performance or economic outcomes. (See Pollitt et al. 1993 for a review.)

Among the most influential studies of the impact of child nutrition on economic well-being is a randomized intervention conducted by the Instituto de Nutrición de Centro América y Panamá (INCAP) in four villages in rural Guatemala between 1969 and 1977. Two villages were randomly assigned to receive a treatment of a fortified drink, Atole, which was high in energy and protein and contained iron and niacin. The other two villages were provided with sugared water, Fresco. Children from birth to age 7, along with pregnant and lactating mothers, were eligible to receive the drink daily at a community center in each village. The study population was poor and undernourished. Prior to the intervention, about four of every five children age 2 in the villages were stunted (height for age z-score < 2), and about one-third exhibited wasting (weight for age z-score < 2).

Children exposed to the treatment are taller at 36 months than children in the same villages who did not receive the treatment during their first few years of life. To isolate the impact of the supplementation, these differences across cohorts are contrasted with differences across the same cohorts in control villages. This is an important result because it establishes that a nutrition intervention in early life can have long-lasting effects on the health and well-being of a population (Martorell et al. 2005).

Maluccio et al. (2006) use a similar approach to measure the impact of nutrition supplementation on cognitive achievement. In a follow-up study conducted in 2002–04, when the study subjects were adults, abstract reasoning skills were measured

with a Raven's Colored Progressive Matrices test. Subjects who were exposed to the treatment during the first three years of life performed better on the test than those not exposed in early life. There is a small impact on females' completed schooling and language skills.

Related research explores the impact of the nutrition supplementation on economic productivity in adulthood. Using the same follow-up survey, Hoddinott et al. (2008) apply the same empirical strategy of comparing the cohort that was fully exposed to the supplement with earlier birth cohorts in the treatment villages and contrasting the difference in economic outcomes with the difference between the same cohorts in the control villages. They find that among males who were working and were fully exposed to the supplementation as a child, hourly earnings were 40 to 60 percent higher than they would otherwise have been. This is an extremely large effect. Elevated productivity was largely offset by reduced hours of work. Annual earnings are estimated to be about 7 percent higher among treated males relative to controls, although this difference is not significant (which may reflect inadequate power because of small sample sizes). Among females, the authors find that treated individuals tend to earn less per hour, work fewer hours, and have lower annual earnings than controls but that none of these differences is significant.

In sum, females benefit more in terms of school performance than males. Among those males who earn income, however, there is a 50 percent premium in hourly earnings for those in the treatment group relative to controls. Compensating reductions in hours of work among these males result in much smaller effects on annual earnings. There are no effects on the labor market outcomes of females. Taken together, the results are puzzling. It is possible that they reflect a complex set of choices related to schooling, marriage, and labor market participation that differs between males and females. It is also possible that the estimated effects are contaminated by attrition that differs between treatments and controls and also between males and females in ways that are related to labor market and schooling outcomes. The authors conclude that this is an unlikely explanation because they find little evidence of systematic differences between those who attrited and those who did not. These comparisons are based on observed characteristics when the respondents were young children. It is possible that there are differences between the two groups that are not observed at baseline, either because they were not measured at that time or because of changes that occurred in the lives of the respondents after the baseline.

Putting attrition aside, the results on the longer-term impact of this early child nutrition intervention suggest that improvements in the nutritional status of children in the first few years of life lead to improved human capital outcomes (as indicated by better health in later life, improved cognitive functioning, and, among females, more schooling) and elevated economic productivity among males in the labor force. Since, however, there is little evidence that annual earnings are higher among treated cohorts, the benefits of improved childhood nutrition do not appear to translate into greater economic prosperity at the aggregate level.

The bulk of the literature on the link between childhood nutrition and economic prosperity in developing countries has, appropriately, focused on undernutrition.

However, in advanced countries, rates of pediatric obesity are high, and the evidence indicates that obesity among young children is an emerging concern in developing countries passing through the nutrition transition (Popkin 1993; de Onis and Blossner 2000). Rapid changes in nutrition levels across cohorts in these countries may presage serious health problems as these children enter midlife. As noted above, the thrifty-gene hypothesis suggests that it is children who are low weight at birth but overweight in adolescence who are at greatest risk of cardiovascular disease and diabetes in later life. Low birthweight is far more common in lower-income countries and is more likely among children whose mothers were poorly nourished as children, as indicated by their height as adults (Victora et al. 2008). There is very little scientific evidence on whether and how childhood obesity affects social and economic success in later life. This is an important gap in the literature.

Although nutrition has played a central role in the literature on health and development, communicable diseases remain a leading cause of death in many low-income settings. Putting aside HIV, many of these diseases are preventable at modest cost. As an example, intestinal helminth infections, or worms, are prevalent in many parts of the developing world and can be effectively treated with readily available antihelminthic drugs. High worm loads result in iron-deficient anemia and protein-energy deficiency, which have, in turn, been associated with reduced cognitive development. The evidence from randomized trials that measure the impact on the cognitive performance of subjects receiving pharmacological interventions to treat helminth infections is inconclusive (Dickson et al. 2000).

Success in treating infectious diseases in a community is likely to have significant externality benefits. Miguel and Kremer (2004) take this as their starting point in estimating the causal effect of treating worms among school-age children, using a random assignment treatment-control design in rural Kenya. In the first year of the study, 25 primary schools were randomly assigned to receive an orally administered antihelminthic drug; those schools and an additional 25 schools received the treatment in the second year, and all 75 primary schools received the treatment in the third year. There are externality benefits: children who were not treated but who lived in the treated areas benefited from the reduced worm loads in those communities. School attendance increased substantially, as indicated by a 25 percent reduction in absenteeism after treatment in treatment areas, relative to controls. Performance on school tests, however, did not respond to the treatment. The authors conjecture that this may reflect the effects of congestion. It may also have to do with what is learned in school and what is tested in examinations—which may itself explain the high levels of absenteeism among primary school children. It is also consistent with there being little effect of helminth infection on cognition (Dickson et al. 2000).

Adult Health and Economic Success

The evidence discussed thus far suggests that health in early life likely affects schooling and economic productivity in later life. We next assess whether changes in health during adulthood affect economic prosperity.

A temporary bout of poor health is likely to be accompanied by reduced time spent working and, possibly, by lower productivity over the short run. Whether these effects translate into longer-term impacts on earnings is less clear. Individuals may compensate by allocating additional time to work when they are in better health. Family members may substitute for a member when he or she is in poor health (Thirumurthy, Graff Zivin, and Goldstein 2008). Chronic diseases and longer bouts of illness are more likely to have permanent effects on economic resources. For example, studies have documented declines in the productivity of individuals working on tea estates in Kenya during the 18 months prior to AIDS-related death or exit from the labor force (Fox et al. 2004).

Determining whether changes in adult health have a causal impact on economic and social success is a substantial challenge. Identification of the health domains that are most important and the magnitudes of their impacts on different dimensions of economic prosperity poses even greater challenges. Convincing evidence will be founded on a model of behavior that takes into account investments in health over the entire life course, expectations about future health and labor market prospects, time allocation in the short and longer runs, and the broader role of family members in these decisions.

Whereas energy supplementation during the first two or three years of life has long-term benefits, experimental studies that involve calorie supplementation of male adults indicate only modest effects on worker productivity. For example, sugarcane cutters in some villages in Guatemala were randomly assigned to receive calorie supplements, while similar workers in other villages were not. There were no differences between the productivity of the workers from the two villages (Immink and Viteri 1981). In contrast, construction workers in Kenya who were randomly given energy supplements were slightly more productive in road digging than the controls (Wolgemuth et al. 1982).

The nutrition literature has highlighted the important role that specific micronutrients play in the etiology of disease and the functioning of the human body. Iron is a good example. Iron deficiency is prevalent throughout the developing world, particularly in South and Southeast Asia, and studies have shown a causal effect of iron deficiency on reduced work capacity, suggesting that there may be a direct link between iron deficiency and economic productivity (Haas and Brownlie 2001).

Iron plays an essential role in oxidative energy production. Iron-deficiency anemia affects physical activity through two main pathways: (a) as hemoglobin levels decline, the maximum amount of oxygen the body can use (aerobic capacity) declines, and (b) as iron stores are depleted, the amount of oxygen available to muscles falls, reducing endurance and causing the heart to work harder for the same activity. Iron deficiency is also associated with, among other effects, greater susceptibility to disease, fatigue, and reduced cognitive development.

Rigorous clinical trials with animals and humans demonstrate a causal relationship between iron deficiency and reduced maximal aerobic capacity (VO_{2max}), which changes by about 25–30 percent as subjects are made anemic or receive adequate iron supplementation. Iron-deficient anemia is also associated with reduced endurance at below maximal work rates.

Demonstrating that iron deficiency impedes maximal capacity and endurance does not reveal the economic consequences of iron deficiency in daily life. Those consequences may be more closely aligned with energy efficiency—the amount of physiological energy required to perform a given task. In fact, evidence from randomized treatment-control studies of Chinese female cotton mill workers and Sri Lankan female tea plantation workers suggest that elevated productivity resulted in changes in time allocation. Specifically, neither study found evidence of greater productivity per hour among iron-deficient subjects who received iron supplements relative to subjects who were not supplemented, but in both cases, treated women reallocated time away from work and toward nonwork activities (Edgerton et al. 1979; Li et al. 1994). Results from a random assignment treatment-control iron supplementation study of male rubber workers, however, indicates that treated individuals who were anemic at the initiation of the study were able to tap about 20 percent more rubber after supplementation, compared with anemic controls (Basta et al. 1979). Interpretation of this result is complicated by the fact that attrition from the study exceeded 50 percent; if those subjects who did not benefit from the study were more likely to attrit, the estimated benefits would be biased upward.

The Work and Iron Status Evaluation (WISE) is designed to examine the immediate and longer-term impact of providing iron supplements to older adults in Central Java, Indonesia (Thomas et al. 2006). In this population, about one-quarter of older men and one-third of older women are iron deficient, as indicated by low hemoglobin. Older males are randomly assigned to receive a weekly iron supplement, or an identical-appearing placebo, for slightly more than a year. Everyone in the man's household is similarly assigned to treatment or control. (In the small fraction of households with more than one older male, a random male is selected to determine assignment to treatment or control for the entire household.)

Respondents who were iron deficient prior to the intervention and who were assigned to the treatment have higher levels of iron in their blood and are able to cycle longer on an ergocycle at the end of the intervention, relative to comparable control subjects. Relative to similar controls, treated subjects are more likely to be working, lose less work time to illness, are more energetic, are better able to conduct physically arduous activities, and enjoy better psychosocial health.

About half the male workers in the study are self-employed (primarily as rice farmers), and the other half are paid a time wage. There is no evidence that hours of work were affected by the treatment, for either time-wage workers or the self-employed. Among males who earned a time wage, there is no evidence of changes in productivity as indicated by hourly earnings. Of course, if their wages are set by an employer, it is not obvious that the worker will reap the benefits of greater productivity. This is not true for the self-employed; males who were iron deficient and self-employed at baseline reported 20 percent higher hourly earnings after six months of supplementation relative to similar controls. Since there was no change in their hours of work, this translates into 20 percent higher income from labor.

Wage workers who received the treatment reduced their amount of time spent sleeping by about 40 minutes per day and reallocated this time to leisure. Self-employed workers made no such adjustments. A picture emerges of iron deficiency

having a causal impact on work capacity and energy needed to complete tasks. The self-employed who benefited from the treatment allocate the additional energy to their fields; they work harder, produce more, and earn more per hour of work. In contrast, wage workers were able to channel their greater energy toward reduced sleep and more time for leisure. After 12 months of supplementation, there is evidence that some of the treated iron-deficient males who were working for a time wage at baseline had shifted to self-employment or had taken an additional job.

WISE demonstrates that iron deficiency has a causal impact on time allocation and economic productivity. It also highlights the importance of taking into account behavioral responses to the experiment itself in assessing the impact of the treatment. Results from the Chinese and Sri Lankan studies reinforce this point; they indicate that iron deficiency had little impact on productivity but did affect how individuals allocated their time.

If iron supplementation improves the health and well-being of individuals, we expect that these people will be more productive not only at work but also in other domains of their lives. Furthermore, they may respond to these improvements in many ways, such as changing the nature of their work and the time spent on work. The results echo the evidence from the INCAP study indicating that treated subjects compensated for increased hourly earnings by allocating less time to work. It is critically important that all these responses are taken into account if we are to understand the impact of health improvements on economic growth. This limits the contributions to the literature of experiments that are designed to isolate specific inputs and outputs in the health and productivity relationship and do not capture the broader ramifications of the health intervention.

Health care costs have skyrocketed and have far outpaced inflation in many countries in recent decades. Much of the increase in advanced economies has been attributed to the development of technologies that improve health and extend life (Cutler 2004). At the same time, calories have become inexpensive relative to other goods (although food prices are subject to considerable variation from year to year and have increased dramatically in recent months). These changes in prices, along with differences in prices across space, provide potentially powerful sources of variation for identifying the impact of changes in the health of adults on social and economic success.

The assumptions under which identification will be valid are not obviously satisfied, at least in general. First, the relative price changes should have no direct impact on the economic prosperity of an individual. In the case of food prices, this rules out the agricultural sector, including all farmers, along with workers in food-related industries. Workers in the health sector and allied industries are similarly affected. Second, variation in prices reflects a combination of a number of factors that conspire to bring demand into line with supply in a general equilibrium. To the extent that those factors also influence economic prosperity, it will be difficult to pin down their impact on health status. This is a special challenge when relying only on temporal or spatial variation in prices. Third, measurement of markers of health in survey data that are affected by variation in prices and also affect economic success raises additional difficulties.

These issues may be sidestepped by relying on experimental or quasi-experimental variation in prices. For example, starting in 1974, subjects enrolled in the RAND Health Insurance Experiment (HIE) were randomly assigned to 1 of 14 fee-for-service health plans. About 1,800 subjects were assigned to a plan that provided free care, and 2,800 subjects were assigned to plans that varied in terms of copayments (from 25 to 95 percent of fees) and total out-of-pocket expenses. Almost three-quarters of the subjects were enrolled in the experiment for three years; the rest participated for five years.

Newhouse and the Insurance Experiment Group (1993) describe the experiment and the main results. Use of health care declined as the price increased, with subjects in the free-care group using health care services most intensively. The evidence on health outcomes is mixed. Newhouse and colleagues conclude that the health of the average person enrolled in the HIE was not adversely affected by reduced use of health services. They note, however, that the health of the poor who were sick was more improved among those who faced lower prices. For example, among the poor who were enrolled in the free-care plan, relative to other plans, mortality was lower, blood pressure improved more, vision was corrected more often, dental health was better, and the incidence of anemia was reduced more. All these indicators were measured through clinical testing or from blood samples.

In stark contrast, the self-reported general health of this group deteriorated during the period! Specifically, of 11 self-reported health indicators, 9 were reported to have worsened among those who received free care relative to those who incurred some cost when seeking health care, although only one of these differences is significant. The clinical and biomarker evidence indicates that these people were, if anything, in better health. One might suspect that, as a result of their greater exposure to the health care system, these people learned more about their health than they would have if they had used services less frequently, and it is this information that is reflected in the worse self-assessed health outcomes. These results provide a practical example of some of the difficulties associated with health measurement in population-based surveys.

Dow et al. (2008) sidestep these problems with measurement of health and examine the impact on labor outcomes of randomly assigned variation in the price of health care. They find that females who received free health care are significantly more likely to be working at the end of the intervention than females who had to pay for health services. The difference in participation rates is about 4 percentage points. Males who had not completed high school are about 8 percentage points more likely to be working if they received free care than if they paid for care (although this effect is only significant at a 10 percent size of test).

A similar experiment conducted in Indonesia involved changes in the prices of health services. To measure the impact of increased user fees, prices of basic services at public health centers were raised in randomly selected treatment districts, while prices were held constant (in real terms) in neighboring control districts. Subjects were interviewed before prices were increased in 1991 and again two years later. Paralleling the results for the HIE, respondents in the areas where prices had increased used less health care and also reported their general health as having improved

relative to subjects in the control areas where prices had not been changed. However, more specific indicators of health indicated that higher prices resulted in worse health. Specifically, respondents in areas where prices increased reported that they had more difficulty doing physically demanding tasks, spent more days away from work, and had more days in bed because of illness. These results further underscore the challenges associated with health measurement.

Turning to work outcomes, in areas where health care prices increased, employment rates rose among males and females relative to those in areas where health prices did not change. These effects are largest among the least educated. The most plausible interpretation of both the HIE and Indonesian results is that the average treatment effects on labor supply indicate a causal effect of improved health on the allocation of time to the labor market.

Evidence regarding the impact of health on economic productivity in the Indonesia study is less clear. Wage growth was lower among workers in the market sector who lived in areas where health care prices were raised, relative to the control areas. The estimated effects are large (about 15 percent) but are only significant at a 10 percent size of test.

Quasi-experimental evidence supports the conclusion that health affects employment outcomes. During the 1960s and early 1970s, Canada introduced national health insurance. Taking advantage of the phased introduction of the system across provinces and occupations, Gruber and Hanratty (1995) find that employment and wages increased as workers were covered by national health insurance. The authors conclude that labor demand rose because workers were more productive, either because of increased job mobility and better matching of skills or because their health improved as a result of being covered by health insurance.

Although the introduction of national health insurance was not designed as an experiment, this study exploits the fact that some people were covered by the system earlier than others. The plausibility of the results rests crucially on the extent to which this “natural experiment” approximates random assignment. The authors provide a compelling argument in favor of this interpretation.

Designing health policy innovations so that they are phased in and rigorously evaluated during the phase-in period will not only contribute to improved policy design and implementation; it also has the potential to facilitate identification of the broader impacts of the policy innovations, beyond health care and health outcomes to economic and social indicators. It is unfortunate that this design is relatively rare.

Turning to the literature based on observational data, the challenge is to identify instruments that are, first, predictive of health status and, second, valid in the sense of not being correlated with unobserved factors that affect the economic or social outcome of interest in the model. The experimental results suggest that prices of health inputs (including foods) may satisfy the first condition for measures of productivity. Labor supply effects are harder to identify in this context because health inputs affect real wages and thus the value of time.

Studies have explored the relationship between body mass index (BMI) and labor market outcomes. BMI, which depends on net energy intake and varies over the life course, captures both longer- and shorter-run dimensions of nutrition. It is related to

VO₂max and thus to aerobic capacity and endurance independent of energy intake (Spurr 1983). Whether this pathway is one through which health influences productivity in an important way is not obvious, since many jobs do not require sustained physical effort. Treadmill tests suggest that excess (fat) weight affects the efficiency at which energy is transferred to work output (Cureton 1992).

Drawing on data on workers in the rural Philippines, Haddad and Bouis (1991) find no effect of BMI on earnings of agricultural workers. Using the same sample, however, Foster and Rosenzweig (1993, 1994) highlight the difficulties of observing health and argue that employers use BMI as a marker for health. They contrast workers paid by time with those paid on a piece-rate basis and find that BMI affects the wages of only time-rate workers. This suggests that it is not health (or at least robustness) that is rewarded in the labor market; rather, there is a benefit from conveying the impression of being in good health. It is not clear why employers would be consistently fooled into paying a worker more because he is heavier (but not more productive) or why employers would not learn about the productivity of workers.

Thomas and Strauss (1997) explore this issue among male and female workers in urban Brazil. With relative food prices as instruments, BMI has a positive impact on the hourly earnings of both employees and the self-employed, suggesting that BMI is more than just an easily observed marker of health used by employers. Since the effect of BMI on hourly earnings is largest among the least educated, who are the most likely to do manual labor, the authors conjecture that BMI may be a marker for strength. Similar evidence is reported for African workers by Glick and Sahn (1998) and Croppenstedt and Muller (2000). There is little empirical evidence relating BMI to labor supply, although BMI has been shown to affect the proportion of working time that is spent on very physically demanding activities by men (Pitt, Rosenzweig, and Hassan 1990; Bhargava 1994; Fafchamps and Quisumbing 1999).

Adams et al. (2003) take a different approach to evaluating the direction of causality underlying the associations between health and SES. They examine whether prior SES predicts current health status, conditional on an extensive set of information on health histories. Under the assumptions of their model, if prior SES does not predict health innovations, then a direct causal pathway from SES to health can be rejected. This is essentially a test for Granger causality. Using data for older adults in the United States, they conclude there is little evidence that SES has a causal impact on health except perhaps in the case of psychosocial health. The authors attribute the latter to the fact that Medicare does not cover mental health services. Their conclusions on the impact of health on SES are more tentative, although they generally find little evidence that health affects wealth of older adults.

The assumptions underlying these tests are not innocuous. The models include a battery of prior health indicators, each of which is the outcome of choices made over the life course. If those choices are correlated with SES, the assumptions underlying the empirical strategy will be violated. In addition, there are legitimate questions about whether the testing strategy has sufficient power to detect causal effects of SES on health in a broader class of models.

In an effort to assess the importance of this concern, Adda, Chandola, and Marmot (2003) conduct a replication using the same empirical methods with data

from the United Kingdom and Sweden. They find results that are remarkably similar to those for the United States, including the evidence on psychosocial health. This is troubling because in the United Kingdom and Sweden, all health services—including mental health services—are available free of charge. The authors conclude that the testing strategy adopted by Adams et al. (2003) lacks power against plausible alternative models.

It is likely that the dynamics underlying observed health and observed SES are very complex. For example, a person's health may be in decline prior to a diagnosis of, say, heart disease, and the decline in health may result in decreased earnings because of reduced hours or lower hourly earnings. Attempts to link the diagnosis to reduced wealth are unlikely to be fruitful. Similarly, a person who perceives that his or her health is in decline and anticipates having reduced earnings in the future may choose to increase his or her labor supply and thus current earnings.

More generally, the dynamic links between health and economic prosperity have been little studied. This is an important gap in the scientific literature, for several reasons. Health is a stock that evolves over time, and prior health behaviors—and health shocks—likely influence current economic status. Virtually nothing is known about the speed with which the effects of health transitions at the individual level are transmitted to the labor market in low-income settings. Does a period of poor health (or a negative health shock) put a worker on a permanently lower wage trajectory, or do the negative consequences of ill health dissipate as health subsequently improves? The extent of catch-up likely depends on the nature of the health problem, the structure of the labor market, and the worker's characteristics, including age, education, and the extent to which the individual has a buffer of resources on which to draw in bad times.

Development of a better understanding of the dynamic interplay between health and economic prosperity is likely to shed new light on some of the mechanisms through which the two covary. Apart from the econometric advantages associated with analyses based on repeated observations of the same person over time, “natural experiments” arising from unanticipated variation in a respondent's life are likely to prove a powerful resource in this literature.

Thomas and Frankenberg (2000) provide an example. The first column of table 2, using data on males age 18–68 interviewed in the 1997 wave of the IFLS, indicates that the elasticity of wages with respect to BMI is 2.0. The results in the second column control for height, age, education, and location of residence: the elasticity is reduced to 1.0, indicating that current weight reflects, in substantial part, human capital and background characteristics. Prior BMI might serve as a control for these characteristics, and accordingly, exploiting the repeat-observation dimension of the IFLS, BMI measured in 1993 is added in column 3. Prior BMI does predict current wages, and it soaks up some of the correlation with current BMI. Controlling 1993 BMI, 1997 BMI can be interpreted as the weight gain since 1993; its effect is also positive. It is not obvious what to make of this correlation. It is possible that increased wages were spent on more energy intake (or less energy output) or that changes in both wages and weight arise because of some other unobserved factor.

To explore this further, we turn to wages measured in 1993. Column 4 essentially replicates the 1997 results reported in column 2. Wages in 1993 are related to BMI

TABLE 2. Dynamic Relationship between Body Mass Index (BMI) and Hourly Earnings of Adult Males, Indonesia Family Life Survey, 1993 and 1997

Covariates	ln(hourly earnings), 1997			ln(hourly earnings), 1993	
	BMI97 only	Plus height and education	BMI97 and BMI93	BMI93	BMI97 and BMI93
	(1)	(2)	(3)	(4)	(5)
ln(BMI) in 1997	2.03 (0.12)	1.03 (0.10)	0.82 (0.16)		0.35 (0.17)
ln(BMI) in 1993			0.34 (0.17)	0.96 (0.14)	0.78 (0.18)
ln(height)		2.36 (0.34)	2.40 (0.44)	2.27 (0.40)	2.52 (0.44)
Years of education (spline)					
0–5		0.05 (0.01)	0.04 (0.01)	0.05 (0.01)	0.05 (0.01)
6–11		0.09 (0.01)	0.10 (0.01)	0.09 (0.01)	0.10 (0.01)
>11		0.11 (0.01)	0.10 (0.01)	0.14 (0.01)	0.14 (0.01)
Age and location controls	No	Yes	Yes	Yes	Yes
R ²	0.06	0.29	0.30	0.30	0.30

Source: Thomas and Frankenberg 2000.

Note: Sample includes adult males age 18–68. Earnings and BMI are measured at time of each survey. Height and education are measured at baseline. Earnings, BMI, and height are specified in logarithms. Numbers in parentheses are standard errors robust to arbitrary forms of heteroskedasticity and take into account correlations resulting from survey clustering. Models 2–5 include age, specified as a linear spline, and location measured by indicators for each province and for urban sector.

in 1993 and to BMI in 1997 (column 5). BMI in 1993 remains significant. However, weight gained between 1993 and 1997 is a significant predictor of wages measured in 1993, prior to that weight change. This is compelling evidence that BMI and wages are jointly determined (or are influenced by other unobserved factors) and that the regressions in table 1 cannot be given a causal interpretation.

This evidence raises further doubts regarding the validity of the approach to identifying causal effects taken by Adams et al. (2003). We conclude that modeling the dynamics underlying the evolution of health and economic prosperity remains an important scientific endeavor.

Conclusions

The weight of the evidence indicates that fetal health and early childhood nutrition have a causal impact on human capital accumulation and economic productivity. There is some evidence indicating that changes in adult health affect labor outcomes, particularly productivity. Although the exact mechanisms underlying these relationships are not entirely clear, the most convincing evidence is distinguished by the coexistence of carefully conducted experimental studies and observational studies that

have documented sizable effects of nutrition on productivity indicators. Explanations that are founded on an understanding of the underlying biological mechanisms have proved to be a powerful force in support of these conclusions.

Little attention has been paid to the behavioral response to longer-term improvements in health and productivity. The limited evidence suggests that short-term increases in productivity have little immediate impact on time allocation but that if the productivity gains persist over the longer term, individuals reduce their hours at work, as would be suggested by theory. If this turns out to be a general result, it is important.

Studies that focus exclusively on productivity and studies that focus solely on annual earnings will provide incomplete descriptions of the relationships between health and labor outcomes. By ignoring the fact that individuals choose to substitute time away from work, these studies will tend to understate the welfare gains associated with health improvements, which are not reflected in aggregate statistics of economic growth. In general, scaling estimated productivity effects to the aggregate level likely will overstate the extent to which the macro association between health and income can be attributed to a causal effect of health on economic prosperity.

A plausible argument can be, and has been, made that other dimensions of health likely affect economic prosperity. These links, however, have not been established as reliably as the links between reducing nutritional deficiencies and productivity. There is little evidence on the impact on socioeconomic success of obesity and related health conditions, of infection and inflammation, or of stress and injuries, despite the prevalence of these health concerns around the globe. And, remarkably little is known about the impact on economic prosperity of psychosocial health problems.

The cumulative effect of multiple health insults on health outcomes in later life has been highlighted in the literature on older adults in advanced economies (McEwen 2000). It has received little attention in the developing-country literature even though health insults are more frequent and potentially more damaging in lower-income societies and may take their toll in early life or midlife. It remains unclear whether wear and tear on the body resulting from health insults have implications for economic prosperity.

More generally, there is a paucity of evidence on the dynamics linking innovations in health and innovations in economic prosperity, particularly over the longer term. Studies that take into account the shorter-term and longer-term behavioral responses that accompany health innovations are few and far between.

In the literature relating health to SES, inadequate attention has been paid to the various domains of SES. The evidence discussed above suggests that the impact of particular health markers on productivity at work and on time allocation may be quite different and that these differences may only emerge over the medium or longer term. Similarly, there are likely to be distinct effects of health on education and cognition in early life and on cognitive performance in later life. The impact of health on productivity in home production, choices between consumption and savings, and intertemporal preferences are also likely to differ.

These are all critical gaps in our knowledge. The task of filling them calls for a sustained program of research that integrates the advantages of randomized treatment-control designs, phased policy interventions, and broad-purpose, longitudinal social

surveys. Each of these tools, on its own, is unlikely to provide the scientific evidence necessary to make progress in the field and thereby influence policy design.

Sustained longitudinal studies that collect repeated observations on the same individual will be necessary for understanding the dynamic aspects of the links between health and SES over the short and longer terms. Broad-purpose panel surveys, in combination with the changes that will occur in the global economy, have the potential to provide a rich laboratory for understanding the links between health and economic success. Financial crises, the collapse of markets, price spikes, and natural disasters are all candidates for this research. It is often difficult, however, to separate shocks from events that might have been predicted, in which case the estimated effects of the observed change will be contaminated by behavioral responses that anticipate the change. The task of separately identifying the effects of multiple, related changes is very complex, and the level of complexity is likely to increase as the magnitude of the shock increases.

Reliance on these natural experiments alone is unlikely to suffice. Pilot studies of the impact of policy changes have the potential to yield important new insights that will improve policy design and affect the state of science. Coordination of longitudinal health and socioeconomic surveys with phased policy changes and health interventions provides opportunities for more planning in the study design and greater control over the variation that people confront.

There are good reasons for expanding the horizons of experimental studies. Specifically, measuring broader socioeconomic outcomes than is typically the case, examining behavioral responses to the treatment, and following subjects for an extended period will substantially increase the contributions of these studies. For example, an intervention that seeks to eradicate malaria in an area might not only measure its incidence before and after the intervention but also track changes in economic productivity and behavioral responses to the intervention, such as changes in work status, hours of work, time allocation, migration, expectations about the future, planning horizons, and investments. Since economic and social changes will probably take some time to emerge, subjects will need to be tracked beyond the intervention period. Taking account of the medium-term and longer-term impacts on both health and economic outcomes will provide a more comprehensive assessment of the intervention.

Randomized interventions or phased trials in parallel with large-scale panel studies have the potential to significantly affect science and policy design in cost-effective ways. A quantum leap in the breadth and quality of research in this area can be achieved by exploiting rapidly evolving technologies for health measurement in a field setting, in conjunction with innovative techniques for combining state-of-the-art survey data and administrative records on individuals, their families, and their communities.

Developing the knowledge base necessary to understand the complex linkages between health and economic prosperity is far from straightforward. Yet the challenges provide an extraordinary opportunity for health scientists, social scientists, and practitioners to collaborate to generate new knowledge that may have a lasting impact on the well-being of populations across the globe.

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